



Internship

(As FullStack Developer)

Project: Online Course & Learning Management System

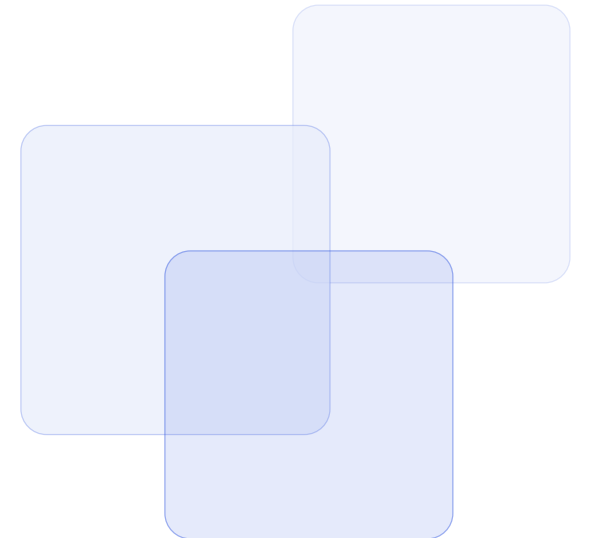
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Batch: **2CSE37**

Program & Semester: **B.Tech & Sem-3**

Organization Name: **CODEC Technologies**



INTRODUCTION

One platform for courses, learners and the people who run them

A web-based Learning Management System built during a Full Stack Developer internship at Codec Technologies.

INSTRUCTORS

Build and publish courses

Upload content, structure a curriculum, and set assessments.

LEARNERS

Enroll and progress

Browse the catalogue, take courses, and see how far they have come.

ADMINS

Oversee the system

Manage users, courses, and access from one dashboard.

PROBLEM STATEMENT

Classroom learning does not scale; the tools that should help are **fragmented**

TODAY

Learning limited by location, schedule and pace

No affordable, unified system to host courses

Learner records and progress kept separately

Existing tools hard to scale or customize

Institutions stitch several tools together.

WHAT IS NEEDED

One web application

Create, manage and deliver courses online

Role-based access for admin, instructor, student

Progress tracked in the same place as the content

This project scopes exactly that.

Project objectives

01

Full-stack LMS with three roles

Admin, instructor and student, each with their own permissions.

04

Responsive interface

An intuitive UI that works across desktop, tablet and phone.

02

Course creation and structure

Content upload and a curriculum organized into modules.

05

Secure, scalable backend

A REST API architecture with validation at the boundary.

03

Authentication and tracking

Login, enrollment, and progress recorded per learner.

Technology Stack



Frontend Layer

React.js & Router: Component-based single-page navigation.

HTML5 & CSS3: Semantic markup and responsive utility styling.

JavaScript (ES6+): Dynamic client interaction and state control.



Backend Layer

Node.js: Scalable server-side JavaScript execution environment.

Express.js: Lightweight REST API routing and middleware framework.

JSON: Standardized client-server data interchange format.



Database & Tools

MongoDB & Mongoose: Document storage with structured schemas.

Git & GitHub: Version control and collaborative code tracking.

Postman & VS Code: Endpoint testing and IDE workflows.

Functional modules



USER

Accounts and access

Registration, login, and role-based access for admin, instructor and student.



COURSE

Create and structure

Course creation, content upload, and curriculum structuring into modules.



ENROLLMENT

Browse and progress

Browsing the catalogue, enrolling in a course, and tracking completion.



ASSESSMENT

Quizzes and grading

Quizzes, learner submissions, and grading tied back to the course.



ADMIN

Oversight dashboard

A single dashboard for user and course oversight across the platform.

Development workflow

BRANCH

Codebase on Git and GitHub — a feature branch per module.

API FIRST

Endpoints built and tested in Postman — before any frontend integration.

IN PARALLEL

Frontend and backend developed side by side — integrated at sprint checkpoints.

REVIEW

Reviewed and merged once the module passes testing — then the next branch opens.

EXPECTED OUTCOME

A working LMS, and **the practice of shipping one**

- **A fully functional LMS.** Students browse, enroll in, and complete courses in one place.
 - **Instructor content tools.** Courses and their material are created and managed without help from an admin.
 - **A centralized admin dashboard.** Users and courses are overseen from a single view.
 - **Hands-on full-stack experience.** API design, database modelling, and real-world project delivery.
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Future enhancements

- 1 **Live virtual classrooms.** Video conferencing integrated into the course page.
- 2 **AI-based course recommendations.** Suggestions tailored to learner progress.
- 3 **Mobile app version.** The same courses, available on the go.
- 4 **Certification and badges.** Issued automatically on course completion.
- 5 **Analytics dashboard.** Deeper insight into learner performance.

Next · live classrooms and certificates build on modules already in place

Later · recommendations and analytics need usage data first

Separate track · the mobile client reuses the same REST API

Thank You